

Homework 2: Metallic materials

Year 9 Science · about 40 minutes

Section A: A Section B: M / E

Section A

1 Write the meaning of each word. **[A]**

(a) density

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(b) alloy

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(c) corrosion

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2 Name the heat treatment. **[A]**

(a) A blacksmith wants a horseshoe to be soft, so it is easy to shape.

(b) The edge of a chisel must be very hard.

3 Write the word equations. **[A]**

(a) lead + oxygen $\rightarrow$ _____

(b) sodium + hydrochloric acid $\rightarrow$ _____

Quick review

4 Name the property of metals that makes each one a good choice. **[A]**

(a) the copper base of a saucepan

(b) gold contacts inside a phone

(c) a steel bridge cable

Section B

5 A gold-coloured crown has a mass of 965 g and a volume of 50.0 cm^3 . The density of pure gold is 19.3 g/cm^3 . Is the crown pure gold? Show your working. **[M]**

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6 Aircraft bodies are made of an aluminium alloy, not steel. Use the data to explain why.

[E]

Material	Density (g/cm ³)	Strength
aluminium alloy	2.8	high
steel	7.8	very high

7 The steel exhaust pipe of a car rusts faster than the steel body of the car. Explain why.

[M]

8 A student adds the same mass of four metals to the same volume of dilute acid and measures the temperature rise.

[E]

Metal	magnesium	zinc	iron	copper
Temperature rise (°C)	18	7	3	0

(a) Put the metals in order of reactivity, most reactive first.

(b) Use the data to explain your order, and name two things the student kept the same.

9 Sam writes about alloys and reactions.

[E]

Sam's answer

1. Steel is harder than pure iron because all the atoms in steel are the same size, so the layers slide easily.
2. magnesium + hydrochloric acid makes magnesium chloride + water
3. Bronze is an alloy, so it is a mixture of metals.
4. When copper is added to dilute hydrochloric acid, bubbles of hydrogen are given off.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

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Also set: Metals booklet p.10–11 (draw the solder graph, then answer Q1–2) and the crossword on p.16.